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Future Skills in the GenAI Era: A Labor Market Classification System Using Kolmogorov–Arnold Networks and Explainable AI

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Abstract: Generative Artificial Intelligence (GenAI) is widely recognized for its profound impact on labor market demand, supply, and skill dynamics. However, due to its transformative nature, GenAI increasingly overlaps with traditional AI roles, blurring boundaries and intensifying the need to reassess workforce competencies. To address this challenge, this paper introduces KANVAS (Kolmogorov–Arnold Network Versatile Algorithmic Solution)—a framework based on Kolmogorov–Arnold Networks (KANs), which utilize B-spline-based, compact, and interpretable neural units—to distinguish between traditional AI roles and emerging GenAI-related positions. The aim of the study is to develop a reliable and interpretable labor market classification system that differentiates these roles using explainable machine learning. Unlike prior studies that emphasize predictive performance, our work is the first to employ KANs as an explanatory tool for labor classification, to reveal how GenAI-related and European Skills, Competences, Qualifications, and Occupations (ESCO)-aligned skills differentially contribute to distinguishing modern from traditional AI job roles. Using raw job vacancy data from two labor market platforms, KANVAS implements a hybrid pipeline combining a state-of-the-art Large Language Model (LLM) with Explainable AI (XAI) techniques, including Shapley Additive Explanations (SHAP), to enhance model transparency. The framework achieves approximately 80% classification consistency between traditional and GenAI-aligned roles, while also identifying the most influential skills contributing to each category. Our findings indicate that GenAI positions prioritize competencies such as prompt engineering and LLM integration, whereas traditional roles emphasize statistical modeling and legacy toolkits. By surfacing these distinctions, the framework offers actionable insights for curriculum design, targeted reskilling programs, and workforce policy development. Overall, KANVAS contributes a novel, interpretable approach to understanding how GenAI reshapes job roles and skill requirements in a rapidly evolving labor market. Finally, the open-source implementation of KANVAS is flexible and well-suited for HR managers and relevant stakeholders.

Keywords: generative AI; skills; Kolmogorov–Arnold Networks; explainable AI; online job advertisements (OJA)



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1. Introduction

Like a tidal force reshaping the shoreline, Generative Artificial Intelligence (GenAI) is redrawing the contours of the global labor market. No longer confined to research labs or

niche applications, GenAI [1–4] has emerged as a transformative wave within the broader arc of Artificial Intelligence (AI) [5], redefining how work is conceptualized, skills are valued, and roles are constructed. GenAI creates new outputs—text, images, or code—based on learned patterns, enabling machines not just to analyze data but to produce content autonomously, thereby reshaping creative and cognitive tasks across industries. As GenAI continues to evolve, it reignites long-standing debates about the relationship between technology and employment. Much like the concerns that arose with the introduction of the assembly line in the early 20th century or mainframe computers in the 1950s and 60s, current discussions on AI reflect a familiar tension between optimism and fear. Optimists argue that automation can liberate workers from repetitive and strenuous tasks, while pessimists warn of widespread job displacement and the erosion of employment security. However, what distinguishes today’s discourse is the shift in focus from manual labor to non-routine, cognitive tasks traditionally associated with white-collar professions. Fueled by breakthroughs in Machine Learning and GenAI, this wave of technological advancement poses new challenges for knowledge workers and unfolds within a highly interconnected global economy—amplifying its potential reach and impact [6].

The origins of AI date back to the post-World War II era, with the Turing Test and John McCarthy’s coining of the term “Artificial Intelligence” in 1956 [7]. Although the 1980s brought an “AI winter,” the rise of Expert Systems marked a focused yet impactful step forward. Subsequent decades saw the emergence of Machine Learning [8,9] and, later, breakthroughs in Deep Learning [10], fueled by increasing access to large-scale datasets [11]. A pivotal moment came with Generative Adversarial Networks in 2014 [12], enabling AI to generate realistic outputs. More recently, transformer-based architectures have transformed natural language processing [13], driving GenAI to the forefront. Yet, its integration into data science workflows highlights persistent challenges: fragmented data sources [14], poor data quality [15], and time-intensive preparation tasks [16]. Communicating complex insights to non-technical audiences remains difficult, and the field’s rapid evolution demands ongoing upskilling and adaptation.

Since late 2022, the rise of generative conversational AI agents—such as OpenAI’s ChatGPT (released with GPT-3.5 in November 2022) and Google Bard (now Gemini)—has drawn widespread attention across professional and social domains [17–19]. ChatGPT surpassed 100 million users within months [20], while Google Bard reached around 30 million monthly visits shortly after launching in March 2023 [21]. In early 2025, DeepSeek, an open-source Chinese Large Language Model (LLM), became the fastest-growing AI application to date, reaching 100 million users in just 20 days without advertising [22]; by 1 February, its daily active users exceeded 30 million. This growth is fueled by major investments from companies like Microsoft, OpenAI, Meta, and Amazon—with Microsoft alone reportedly investing \$10 billion into ChatGPT [23]. These tools are valued for producing human-like content across text, images, audio, and video, offering rapid, personalized access to information. As they integrate into daily workflows, they provoke essential questions about the transformation of roles in fields such as data science. Despite the rise of GenAI, demand for data scientists remains strong, with median salaries exceeding \$100,000 and top earners surpassing \$190,000 [24]. Employment is projected to grow by 36% from 2023 to 2033, adding over 73,000 new jobs and generating around 20,800 openings annually [25,26].

Crucially, the transition to GenAI presents both opportunities and challenges. It boosts automation and analytics but raises concerns around data privacy, ethics, and algorithmic bias. Integrating GenAI demands infrastructure upgrades and workforce training. Additionally, the “black box” nature of these models complicates transparency and trust [27]. As such, GenAI adoption requires rethinking data science practices, prioritizing

ethical awareness, adaptable skills, and interdisciplinary collaboration. Figure 1 visualizes this shift, emphasizing disruption, skill redefinition, and workforce adaptation.

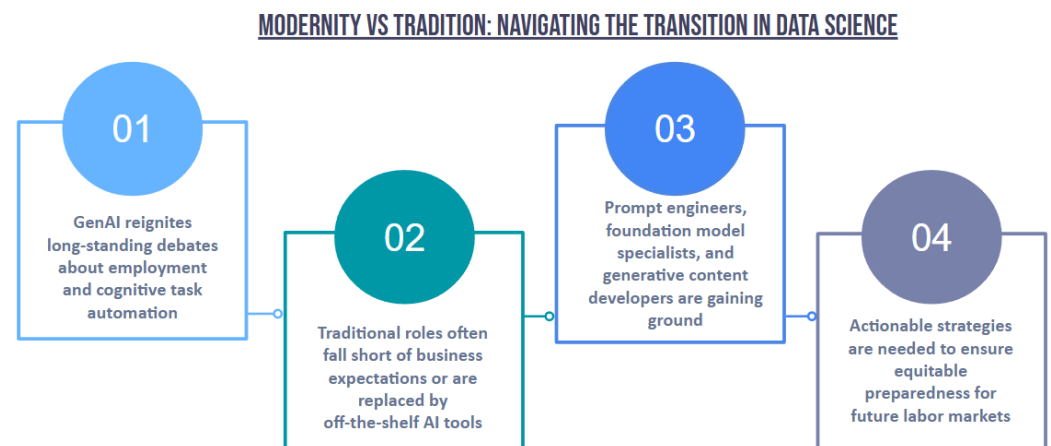


Figure 1. GenAI Transition flowchart.

Crucially, the transition to GenAI presents both opportunities and challenges. It enhances automation and analytics but raises concerns around data privacy, ethics, and bias. Integrating GenAI demands infrastructure upgrades and workforce training. Additionally, the “black box” nature of these models complicates transparency and trust [27]. Thus, GenAI adoption requires rethinking practices, prioritizing ethical awareness, adaptable skills, and interdisciplinary collaboration. Figure 1 visualizes this shift, emphasizing disruption, skill redefinition, and workforce adaptation.

This shift toward GenAI technologies has placed the traditional role of the data scientist under scrutiny [28], as industry demands favor faster, more impactful outcomes. Even when data scientists have strong technical skills, many projects fail to create value because the models are never actually deployed. Complex models that succeed in academic settings often falter with real-world data, leading to inefficiencies. As off-the-shelf APIs and SaaS tools improve, businesses increasingly prioritize speed-to-market over custom model development. In fact, earlier waves of automation and AI—from the 1980s to early 2000s—similarly triggered major task reclassifications across sectors like finance and manufacturing [29,30]. More recently, hybrid roles combining domain expertise and algorithmic fluency have emerged [31,32]. In many current cases, machine learning may not even be required, raising questions about the continued relevance of traditional AI roles, often referred to as narrow or weak AI [33]—focuses on executing a specific task with intelligence.

Although many studies within the GenAI literature explore critical issues such as bias [34], hallucinations [35], and ethical challenges in applied domains like education [36], they tend to focus on specific applications or outcomes rather than the broader algorithmic foundations of GenAI systems. Even studies addressing GenAI’s role in academic tasks (e.g., literature review generation) [37] often do not offer a structured view of the algorithm types themselves, their evolution, or cross-domain patterns. This gap becomes more pronounced when considering the increasing integration of GenAI into the workforce [38], where questions of algorithmic competence and human–machine role adaptation remain underexplored. Despite this growing integration, the literature lacks a comprehensive perspective on how algorithmic advancements translate into evolving skill demands—particularly the transition from traditional data science profiles to GenAI-oriented roles [2].

The lack of clear identification and understanding of labor market demand for these evolving roles delays the effective implementation of workforce strategies and reskilling programs. This paper addresses this critical gap by introducing KANVAS

(Kolmogorov–Arnold Network Versatile Algorithmic Solution), a customized AI algorithm designed to distinguish between the core skills required for traditional AI positions—rooted in statistics, classical machine learning, and data analytics—and those essential for modern GenAI roles such as prompt engineering and AI system orchestration.

In its first phase, KANVAS targets Online Job Advertisements (OJA) that include both traditional AI and GenAI-related competencies, aiming to distinguish them by skill profiles. This distinction emerged in society more prominently following the widespread adoption of LLMs, particularly after the release of ChatGPT. Several studies have presented this dichotomy clearly. For example, Khalil et al. [39] discuss the role of modern cybersecurity in both the generative AI and traditional AI eras; ARTiBA [40] outlines the key differences between these two categories; a study by the University of Illinois [41] explores and contrasts the two concepts; MIT Curve [42] examines the broader shift from traditional AI to generative AI; and Aggarwal et al. [43] describe how traditional AI relies on explicit, rule-based expert systems for domain-specific tasks, whereas modern AI employs data-driven methods such as deep learning to enable adaptive, scalable, and complex problem-solving capabilities. For simplicity, we refer to traditional AI roles as *traditional* and GenAI-related roles as *modern* throughout this study. Data were collected from LinkedIn and Kariera, focusing on roles within the European Union—an active recruitment hub reflecting current employer needs. Skills were extracted using automated tools aligned with the European Skills, Competences, Qualifications, and Occupations (ESCO) [44] and the Lightcast taxonomy, enabling systematic role classification. ESCO taxonomy—developed by the European Commission to identify, describe, and classify professional occupations and skills relevant to the EU labor market and education sectors [45]—comprises over 3000 occupations and 13,900 competences, while Lightcast offers real-time labor market data analytics by aggregating and analyzing millions of online job postings to capture emerging skill trends and evolving occupational demands. To analyze these structures, we apply Kolmogorov–Arnold Networks (KANs) [46], which decompose multivariate functions into interpretable univariate components. Unlike earlier predictive applications, the focus here is on the explanatory power of KANs—highlighting the skills that drive classification into traditional or GenAI roles. This methodological design supports the investigation into GenAI’s workforce impact, guided by four Research Questions (RQ) concerning the evolving skill landscape.

- **RQ₁:** Can KANs accurately and transparently distinguish between modern GenAI roles and traditional AI roles based solely on skill profiles?
- **RQ₂:** How is traditional AI impacted by GenAI and which skills are mainly affected?
- **RQ₃:** Can traditional AI roles be distinguished from their GenAI counterparts?
- **RQ₄:** How can real-time skill explainability be used to support stakeholders in identifying GenAI readiness and guiding upskilling decisions?

In addition, a LLM is employed to classify OJA into the two categories—traditional AI and GenAI—based on the linguistic and contextual cues within each description. This enables a deeper understanding of the competencies needed for the future of work and highlights key areas for policy, education, and workforce development. The main contributions of this study are as follows:

- We introduce KANVAS, the first KAN-based framework for distinguishing traditional and GenAI-related job roles using interpretable, skill-driven classification.
- Our approach leverages Explainable AI (XAI) techniques to identify the most influential skills differentiating modern and traditional AI roles, based on large-scale job postings.
- KANVAS is validated on real-world job advertisements, offering actionable insights for workforce planning and upskilling strategies.

- The trained KAN models enable augmentation of traditional AI job profiles with GenAI skills, illustrating their added value in modern data science.
- Our implementation is publicly available at <https://github.com/dkavargy/KANVAS> (accessed on 19 August 2025).

The remainder of this paper is organized as follows. Section 2 reviews background literature on traditional AI and GenAI applications. Section 3 outlines the KANVAS pipeline, including data collection, skill extraction, and classification. Section 4 presents the results of model evaluation and skill differentiation. Section 5 covers practical applications, while Section 6 explores implications for labor market dynamics. Finally, Section 7 summarizes the key contributions and suggests future research directions.

2. Related Work

KANs represent a reimagining of neural architecture by replacing static linear weights with dynamically learnable univariate spline functions. Introduced by Liu et al. [46], this design leverages the Kolmogorov–Arnold representation theorem to combine strong approximation capabilities with interpretability. Unlike traditional MLPs, KANs balance accuracy and computational efficiency, enabling advances across domains: time-series modeling [47], medical imaging diagnostics [48], and visual classification pipelines [49]. This architectural versatility, along with support for symbolic regression, makes KANs well-suited for high-stakes applications where transparency is essential.

Concurrently, GenAI is driving sector-wide disruption through cognitive task automation, fundamentally reshaping labor markets. In education, systems like ChatGPT automate grading and lesson planning [50]; in healthcare, GenAI generates synthetic medical imagery and diagnostic reports [51]; while in creative industries, text-to-media tools displace traditional workflows. This acceleration has exposed critical research gaps regarding workforce impacts. Current studies inadequately address three systemic consequences: (1) Skill polarization favoring hybrid competencies (e.g., “prompt engineering” over Python (currently version 3.x) scripting) [52]; (2) Temporal misalignment where taxonomies lag behind emergent roles like “LLM hallucination mitigation specialists”; and (3) Geographic fragmentation with innovation hubs exhibiting adoption rates up to three times higher than those of peripheral regions [53]. These dynamics create urgent challenges for workforce development systems ill-equipped to map rapidly evolving skill landscapes.

In parallel with GenAI’s disruptive influence across sectors, recent studies have started to examine its impact on workforce transformation. Research highlights that ESCO and O*NET struggle to keep pace with emerging GenAI-related competencies, often missing skills not yet included in formal taxonomies [54,55]. Few-shot and in-context learning methods have shown promise in extracting complex job-skill patterns beyond rule-based systems [56,57]. Job-aware models like JobFormer use transformer-based architectures to map job descriptions to user profiles, though they still face limitations in handling longitudinal and cross-domain dynamics [58,59]. Additionally, policy frameworks stress the importance of interoperable taxonomies and API-based data flows to sustain adaptive, skill-based labor ecosystems [60].

Traditional job recommendation systems are increasingly insufficient in addressing rapid technological change. Keyword-matching approaches [61] often misinterpret semantically similar skills with different applications—e.g., PyTorch (initial release in 2016; major version 2.0 released in March 2023) for classical ML vs. GenAI. Collaborative filtering models reinforce outdated role-skill links. While NLP-based systems improve skill extraction [62], key issues remain: lack of contextual skill grouping, poor modeling of obsolescence (e.g., manual features), and limited interpretability for policy use [63]. Moreover, many overlook emergent, taxonomy-external skills not captured by formal classifications—leading to alignment blind

spots. A systematic review [64] confirms that traditional filters often miss novel competencies required in today’s labor market. Temporal dynamics of skills—how they evolve—are rarely modeled. Dawson et al. [59] address this through a skill-driven job recommender leveraging longitudinal trends to guide future-proof transitions.

Most critically, no existing framework captures the nonlinear skill adjacencies and transition pathways emerging at GenAI innovation velocities. This triple challenge—KANs’ interpretable function approximation, GenAI’s labor market transformation, and recommendation systems’ contextual rigidity—motivates our integrated solution: A KAN-enhanced framework for real-time skill decomposition and transition mapping in dynamically evolving labor ecosystems.

In summary, while all the above prior studies provide significant advances in understanding GenAI’s algorithmic evolution and workforce implications, they often remain fragmented—either focusing narrowly on technical architectures (e.g., KANs), or offering descriptive insights into labor market impacts without linking them to the algorithmic shifts driving those changes. Furthermore, current recommender and classification systems lack mechanisms for identifying emergent, taxonomy-external skills, and rarely integrate temporal dynamics at scale. These gaps motivate our approach: a large-scale, cross-domain analysis of GenAI algorithms, coupled with a taxonomy-aware model for skill evolution and alignment, aimed at bridging the disconnect between algorithmic innovation and labor market adaptation.

3. Methodological Framework

Inspired by scenario typology methods in futures research [65], KANVAS restructures the classical three-phase approach into a four-phase empirical pipeline for classifying GenAI-related job roles (Figure 2). First, large-scale OJA are collected and preprocessed to extract standardized skill tags using ESCO and GenAI-specific lexicons (Section 3.1). Second, job roles are labeled as modern or traditional using a refined prompt applied to the llama3:8b model (Section 3.2). Third, a KAN is trained on multi-hot encoded skills, with SHAP applied to reveal feature importance (Section 3.3). Finally, classification insights are interpreted to inform curriculum development and AI-readiness strategies, closing the feedback loop (Section 3.4).

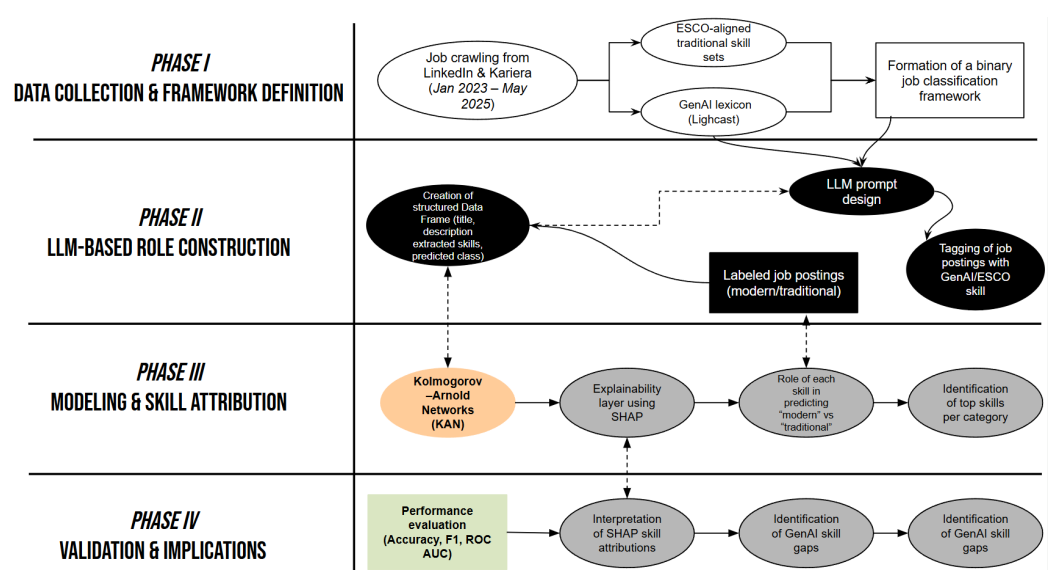


Figure 2. The four-phase job classification framework of KANVAS.

3.1. Data Collection and Framework Definition

OJAs offer valuable insights into current labor market demands, particularly in dynamic fields like data science. As the first phase of the customized algorithm, the focus was placed on capturing both traditional AI and GenAI-oriented roles by collecting postings related to (i) conventional data science tasks—such as data analytics, modeling, wrangling, statistical analysis, and machine learning—and (ii) emerging GenAI skills, including prompt engineering, LLM use, and ChatGPT functionalities. To ensure broad, multilingual, and geographically diverse coverage, two complementary platforms were selected: *LinkedIn* (<https://www.linkedin.com> (accessed on 20 May 2025)), a globally recognized professional network [66] with strong representation in high-skill digital sectors [67], and the *Kariera Group* (<https://www.karieragroup.com/> (accessed on 21 May 2025)), founded in 1997, which is the leading job portal in Greece and became part of CareerBuilder’s global network in 2007 [68], thereby expanding its reach to international candidates and employers. The Kariera Group included platforms such as brightminds, jobbguru, jobmedic, jobs.de, kariera.gr, and others, covering both generalist and sector-specific postings. This combination allowed us to access rich, real-time occupational data reflecting both international and localized trends [69], ensuring our dataset was representative of current labor market dynamics across diverse contexts.

Due to the multilingual nature of European postings, an automated translation step was applied using the *GoogleTranslator* module from the *deep_translator* Python library (<https://pypi.org/project/deep-translator/> (accessed on 20 May 2025)) to convert all job titles and descriptions into English. This ensured linguistic consistency while preserving semantic content. Each job entry included a title and a description detailing role requirements, responsibilities, location, salary, and other contractual details (see Table 1).

Table 1. Sample job advertisement.

Field	Value
Job Title	Developer of Generative AI solutions F/M
Company	XXXX
Location	Clermont-Ferrand, France
Domain	lesjeudis
Date	12 May 2025
Seniority Level	Mid-Senior level
Salary Range	€55,000–€70,000 annually
Contract Type	Permanent
Description	As a GenAI developer in XXXX you must possible and technical expertise to provide Solution Architecture and Design: Define the architecture of enterprise-grade AI applications, ensuring scalability, security, and maintainability. The employee will guide the design of RAG-based pipelines, AI agents and API-driven AI solutions. Optimize prompt engineering and LLM interactions to improve response accuracy & relevance. Ensure LLM security best practices, data privacy... (cont’d)

To support the development of the KANVAS algorithm, two separate search queries were designed to collect OJA reflecting both traditional and modern (GenAI-oriented) data science roles. The goal was to capture a diverse yet focused set of occupations aligned with different skill paradigms. After experimenting with multiple formulations, the final keyword queries were selected as follows:

Traditional AI OJA: *machine learning OR natural language preprocessing OR regression models OR data mining OR predictive analytics OR feature engineering OR model evaluation OR data wrangling OR recommendation systems OR Python OR facial recognition*

GenAI OJA: *ChatGPT OR prompt engineering OR generative AI OR LLM OR transformer (machine learning model) OR deep learning OR convolutional neural networks OR natural language generation OR autoencoders OR reinforcement learning OR AI Agents*

A custom Python-based web crawler was implemented using these queries to extract OJA. The crawler first gathered URLs of OJA and then stored the full content—title, description, company, location, and metadata—into a structured CSV file. The final dataset comprised 9357 OJA collected across several EU countries from January 2023 to May 2025. The dataset was manually checked for duplicates, and no duplication was detected, so no deduplication process was applied. All OJA were collected exclusively from publicly accessible sources (LinkedIn and Kariera job boards), containing no personally identifiable information (PII). The data collection process adhered strictly to the platforms' terms of service and complied with GDPR requirements. The dataset was used solely for academic research purposes and anonymized at all stages of processing, thereby safeguarding employer and candidate privacy. More detailed information regarding the keyword formulation, pilot testing, and manual validation procedures used to construct the traditional and GenAI occupational queries is provided in the Supplementary Materials—Section S3.

Skill Extraction from OJA

The next step in the methodological framework involved the extraction of skills from the retrieved OJA. The rationale behind this decision stems from the fact that job descriptions, though rich in role-specific information, often present heterogeneous and loosely defined skill requirements. Therefore, a structured mapping of job descriptions to formalized skill sets is essential to enable systematic analysis and ensure a consistent semantic framework. A hybrid extraction strategy tailored to traditional AI and GenAI-related job roles was employed to achieve this objective. Traditional AI postings were processed using the ESCOX tool [70], a skill and occupation extractor aligned with the ESCO taxonomy, while GenAI postings were analyzed using a curated lexicon derived from Lightcast's open skills taxonomy. This lexicon (Table A1) comprised 226 manually validated GenAI-related skills. The overall extraction involved exact keyword matching and semantic similarity techniques based on sentence embeddings.

For a detailed description of the extraction pipeline—including examples, tool specifications, and the hybrid algorithm used—please refer to the Supplementary Materials (Section S1—Extraction with ESCOX, Section S2—Pre-processing of the skills lexicon (Lightcast taxonomy), Section S3—Sequential keyword matching).

3.2. LLM-Based Role Construction

While OJA were collected using keywords related to both traditional AI and GenAI, many roles lacked clear distinctions in their titles or descriptions—e.g., “AI Engineer” could imply either statistical modeling or LLM-based tasks. To address this ambiguity, and as part of the second phase of our pipeline, we employed the instruction-tuned llama3-8b model [71], chosen for its strong semantic classification performance and top ranking on the Open LLM Leaderboard [72]. Specifically, the llama3-8b model was selected for its strong performance in practical NLP tasks relevant to job classification and skill recognition. However, as discussed in Section 4.1, llama3-8b was qualitatively validated against other models and achieved higher accuracy scores. In addition and according to recent benchmarking, llama3-8b slightly outperforms both GPT-3.5 and GPT-4 in several classification tasks, including sentiment analysis (https://medium.com/@zljohn18_71393/from-gpt-3-5-vs-

[gpt-4-vs-llama-3-performance-and-cost-analysis-on-practical-nlp-tasks-b1b422a2567d](#) (accessed on 21 May 2025)). While its inference speed is marginally slower, this is not a limitation in our case, as experiments were executed efficiently within a Google Colab Pro environment, as detailed in Section 4. Using a custom prompt (see Listing 1), the model classified job roles as either modern (GenAI-related) or traditional (ESCO-based). This enabled consistent tagging of postings that included GenAI skills (e.g., prompt engineering) versus conventional skills (e.g., regression, BI tools). The resulting labels were treated as benchmarking annotations [73], supported by evidence from Törnberg [74], who showed that LLMs could outperform experts in annotation accuracy. The final dataset—comprised job metadata, skills, and predicted labels—forms the analytical foundation (see Section S4 of the Supplementary Materials).

Listing 1. Prompt generated using a LLM to guide classification decisions.

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“‘You are a highly specialized labor market expert trained to classify job
  roles based on current AI and data trends.
Your task is to analyze the job title and description provided, and
  classify the job into one of two distinct categories:

1. 'modern': Roles that involve recent advances in artificial intelligence
  and generative technologies.
Examples include work with large language models (LLMs), GenAI, prompt
  engineering, RAG (Retrieval-Augmented Generation),
LangChain, CrewAI, synthetic data generation, vector databases (e.g., FAISS
  , Pinecone), diffusion models,
multimodal AI, transformers, embeddings, fine-tuning of models, and modern
  MLOps or LLMops practices.

2. 'traditional': Roles that rely on established methods in data science
  and machine learning, such as regression, classification,
clustering, feature engineering, statistical analysis, BI/reporting, and
  traditional ML frameworks like scikit-learn, TensorFlow (classic usage)
  ,
and general analytics. These jobs do NOT involve GenAI or advanced
  generative AI components.

Only reply with one word: modern or traditional. No explanation. No
  punctuation. No formatting”’

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3.3. Modeling & Skill Attribution

In the third phase of the pipeline, KANs were employed to classify job roles as either modern or traditional, based on multi-hot encoded skill vectors extracted from OJA. KANs were introduced as a novel class of neural networks grounded in the Kolmogorov–Arnold Representation Theorem. This theorem states that any multivariate continuous function can be decomposed into a finite composition of univariate functions and additions. Instead of relying on static activation functions as in traditional MLPs, KANs utilize learnable univariate transformations (typically via B-splines) along the edges, making the model both powerful and interpretable.

$$f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \varphi_{q,p}(x_p) \right) \quad (1)$$

In our study, while the primary objective of the KAN is classification, its architecture also enhances interpretability by allowing us to examine how specific input skills $x_p \in \mathbb{R}$ contribute to the final decision. Each input skill is processed through a learnable univariate function $\varphi_{q,p}(x_p)$, and the results are aggregated via summation. The intermediate values are then passed through a second learnable function $\Phi_q(\cdot)$, producing the final output as defined in Equation (1). This formulation allows us to interpret each skill’s influence

by examining the shape and weight of the univariate transformations, thereby providing actionable insights into which competencies are most indicative of modern GenAI roles versus traditional ones.

To ensure transparency and reproducibility, the study evaluated a range of hyperparameter configurations by varying network width, grid resolution, and learning rate. Based on empirical evaluation, the final setup—with a width of 16, spline grid resolution of 3, and learning rate of 5×10^{-4} —offered the best trade-off between convergence stability, classification performance, and model interpretability. A stratified 80/20 split was applied to maintain class balance during training and evaluation, and a weighted random sampler was used to mitigate label imbalance during training iterations.

For a detailed description of the Kolmogorov–Arnold Theorem, KAN architecture, and the underlying mathematical formulation, please refer to Section S5—of the Supplementary Materials.

3.4. Validation & Implications

After training the KAN, classification performance was evaluated using a set of standard metrics derived from the model's predictions on the test set. The evaluation included accuracy, precision, recall, and F1-score, which collectively offer a comprehensive understanding of both general correctness and class-wise performance.

The use of XAI techniques played a critical role in increasing the transparency of skill-based classification decisions. As AI models such as the KAN increase in complexity, their decision-making processes can become difficult to interpret, resembling a “black box” [75]. XAI provides a means to clarify how specific input features—such as job-related skills—contribute to a model's prediction. This transparency was particularly important when distinguishing between traditional and GenAI-related OJA, as it allows researchers, educators, and policymakers to trust and critically evaluate model outcomes, ultimately supporting more informed decisions in areas such as curriculum development and workforce planning [76].

The Shapley Additive Explanations (SHAP) method is an explainability technique grounded in cooperative game theory [77], designed to attribute a precise contribution to each input feature in a model's prediction. These contributions, referred to as SHAP values, quantified the individual impact of each feature on the predicted outcome for a specific instance. SHAP supports both global interpretability—by summarizing the overall influence of features across a dataset—and local interpretability, by explaining feature effects on individual predictions. A key strength of SHAP lies in its consistency: the sum of all feature contributions corresponds exactly to the difference between the model's output for a given instance and the expected prediction [78]. Despite its interpretive richness, SHAP can be computationally demanding [79], especially when applied to large-scale or highly complex models.

Compared to alternative XAI techniques such as LIME [80] or Integrated Gradients [81], which are primarily used for explaining deep neural networks, SHAP was selected for its strong theoretical guarantees (e.g., local accuracy, consistency) [82] and its unified framework applicable to a wide range of model architectures [83]. Unlike attention weights or gradient-based methods, SHAP enables additive feature attribution, which aligns closely with the need to identify cumulative skill importance across both traditional and GenAI-oriented roles.

For additional implementation details, including sampling strategy and visualization types used, please refer to Section S5.3 of the Supplementary Materials.

4. Results

This section presents the results of KANVAS, conducted in a specialized environment to address the research questions. All experiments and code development were conducted in the Google Colab Pro environment, configured with an *NVIDIA A100 GPU* and *83 GB* of RAM. Section 4.1 highlights the performance of the KAN model and demonstrates its effectiveness in accurately classifying OJA into modern and traditional roles. Section 4.2 focuses on the significance of GenAI-related skills and contrasts them with those found in traditional roles. Section 4.3 presents the KAN-based classification outcomes specifically for modern job roles, while Section 4.4 provides a corresponding analysis for traditional roles.

4.1. Evaluating the KAN Model

In the initial phase, the dataset utilized in this study comprised job postings classified into two distinct categories: modern and traditional roles, based on their associated skill profiles. The full dataset contains 9357 annotated entries, with 5473 postings labeled as traditional and 3884 as modern. Based on the annotations produced by the LLM classifier (serving as the benchmarking annotations), the dataset exhibits a class distribution in which traditional roles constitute approximately 58.49%, while modern roles account for the remaining 41.51%. This classification directly informs our analysis in RQ_1 .

As part of a reliability assessment of the LLM-generated classifications, we explicitly conducted a human annotation protocol [84] to establish a robust gold standard, defined as manually annotated collections of text by human experts. From the complete set of job postings (9357), we randomly sampled 100 descriptions, as creating a gold standard for the entire dataset would be prohibitively costly [85]. These were independently annotated by the first three authors of this study. Each annotator assigned a binary label—either modern or traditional—following the same classification guidelines provided to the LLMs. Agreement among all three annotators was required for a label to be retained. In cases where full agreement was not reached, the posting was replaced with another randomly sampled one, ensuring that the final annotated set contained only unanimously agreed labels (see Annotated sample (<https://docs.google.com/spreadsheets/d/1T2RUdynXSYSptC5QPfZU3rr9BArHgU-5yjai5QJXrNs/edit?usp=sharing>, accessed on 19 August 2025)) and the corresponding llama3:8b outputs on this same set (llama3:8b predictions (https://drive.google.com/file/d/1oRi_0QfMCjgM59kphedUupfSVVY3f6Zp, accessed on 19 August 2025)).

This human-validated dataset served as the gold standard for model evaluation. We compared the predictions of the llama3:8b model with those of four additional LLMs available in the Ollama framework. Accuracy was computed as the percentage of LLM classifications matching the human gold standard. The results (Table 2) demonstrate that llama3:8b achieved the highest accuracy (83%), outperforming all other evaluated models, including smaller reasoning-oriented and instruction-tuned variants. This finding reinforces the validity of our labeling approach and demonstrates that the chosen LLM configuration yields superior agreement with human judgment. In this case, given that llama3:8b achieved high accuracy against the human-annotated dataset, the LLM's classification results can be regarded as a reliable benchmark reference for the remaining automated labeling tasks, with an estimated quantifiable LLM label noise of approximately 17%.

The classification performance of our model—trained to distinguish between *modern* and *traditional* job roles using skill profiles—was evaluated using standard confusion matrix metrics. An 80:20 train-test split was applied to ensure that the results reflect generalization on unseen data. As shown in Table 3, the KAN model achieved 79% accuracy on the 1872-sample test set. A performance asymmetry is observed: traditional roles show high precision (0.89) but lower recall (0.68), indicating reliable positive predictions yet missed

instances. In contrast, modern roles yield high recall (0.88), capturing most relevant cases, though with lower precision (0.66), reflecting more false positives. This suggests differing skill signal distributions across classes. The macro F1-score stands at 0.77, indicating balanced classification. SHAP-based interpretability findings follow in the next section.

Table 2. Comparison of LLM classification accuracy against human-annotated gold standard (100 job postings).

Model	Size (GB)	Accuracy (%)	Error (%)
deepseek-r1:1.5b	1.1	79	21
deepseek-r1:8b	5.2	80	20
mistral:instruct	4.1	78	22
deepseek-r1:latest	5.2	82	18
llama3:8b (Ours)	4.7	83	17

Note: Bold values indicate the best-performing model in the evaluation.

Table 3. Classification performance of the KAN model.

Class	Precision	Recall	F1-Score	Support
Modern	0.67	0.88	0.76	777
Traditional	0.90	0.68	0.77	1095
Accuracy		0.79 (on 1872 samples)		
Macro Avg	0.78	0.78	0.78	1872
Weighted Avg	0.80	0.78	0.78	1872

Model Benchmarking Details. To validate the suitability of KANs, we conducted a benchmarking experiment against seven conventional classifiers. All models were trained on identical skill-encoded features derived from job postings labeled as either “traditional” or “modern” roles. The evaluation focused on Accuracy, Precision, Recall, F1 Score, and ROC AUC, as summarized in Table 4. KANs marginally outperformed all models in terms of ROC AUC and validation accuracy, supporting their selection for this classification task. Notably, models such as SVM and Random Forest came close in performance but lacked the smooth interpretability and interpolation capabilities that KANs afford when used with SHAP explanations.

Table 4. Performance Comparison of Classical Models on Job Type Classification.

Model	Accuracy	F1 Score	Precision	Recall	ROC AUC
Logistic Regression	0.7607	0.8040	0.7716	0.8393	0.8292
Decision Tree	0.7838	0.8258	0.8162	0.8356	0.8379
Random Forest	0.7921	0.8515	0.8319	0.8721	0.8861
Naive Bayes	0.7329	0.7784	0.7562	0.8018	0.8107
K-Nearest Neighbors	0.7826	0.8236	0.7838	0.8676	0.8441
SVM	0.7943	0.8318	0.7973	0.8694	0.8428
Gradient Boosting	0.7730	0.8140	0.7815	0.8493	0.8462
KANs (Ours)	0.7960	0.8532	0.8384	0.8695	0.8885

Note: Bold values indicate the best-performing model across all evaluation metrics.

The second validation step aimed to further evaluate the performance of KANs by implementing a simple rule-based classifier using a curated list of indicative GenAI and traditional AI keywords. While the method achieved a high coverage of 97%, its accuracy was only 44.7%, due to a strong bias toward detecting modern job signals. The rule-based model correctly identified all modern jobs (recall = 1.00) but failed to detect traditional roles

effectively (recall = 0.04), highlighting the limitations of using heuristics in this context and reinforcing the need for data-driven, generalizable models. More information is provided in the Supplementary Materials Section S6.2.

After the two validation steps—which demonstrated that KANs outperform other conventional models and that LLMs can serve as a reliable benchmark annotation in this context—we observe that Figure 3 illustrates the training and validation accuracy of our KAN-based classification model over 100 training epochs. Initially, both accuracies exhibit a steep increase, indicating rapid learning. The training accuracy continues to rise and converges around 79%, while the validation accuracy stabilizes slightly below that level, around 76%. This consistent gap between training and validation performance, without a significant drop in the latter, suggests that the model has achieved a good generalization capability without overfitting. This moderate but stable accuracy indicated the robustness of the KAN framework in distinguishing between modern and traditional job roles based solely on skill-based input features.

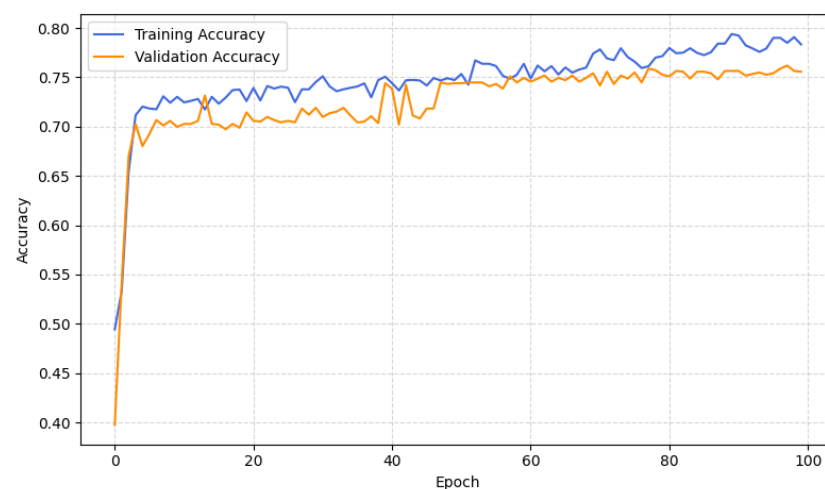


Figure 3. KANs accuracy over epochs.

While standard metrics such as accuracy, precision, and F1-score quantify model performance, they offer limited insight into the underlying features driving predictions. In this context, KANs provide an interpretable framework through their use of learnable univariate functions. When combined with SHAP-based attribution, this architecture enables detailed analysis of which skills contribute most to the classification of roles as modern or traditional.

As illustrated in Figure 4, the ROC curve demonstrated the discriminative capability of our proposed KAN-based framework in classifying *modern* and *traditional* job roles. The area under the curve (AUC) reaches 0.86, reflecting strong classification performance. The ROC curve showed a strong true positive rate at low false positive rates, indicating a reliable balance between sensitivity and specificity.

Beyond overall classification accuracy, a skill-level analysis was performed to identify the competencies that most influence the assignment of job roles as modern or traditional. This interpretability layer highlights how specific skill features contribute to model decisions and supports human-understandable insights into GenAI-related role differentiation. For further details regarding the job classification methodology, please refer to Supplementary Material—Section S6.1. Statistical validation of the distribution of the two skill categories is provided in Sections S6.2 and S6.3, respectively.

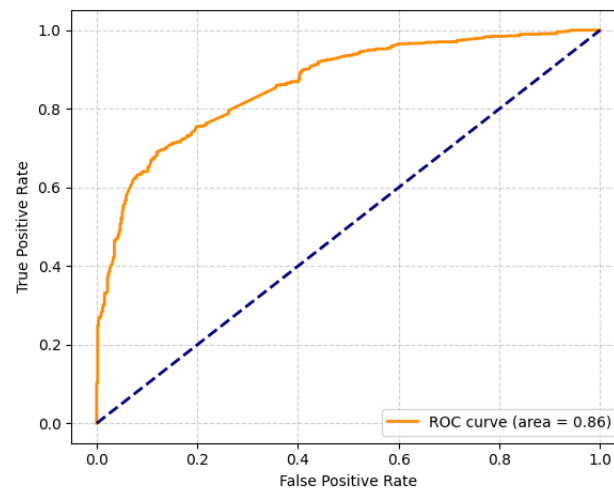


Figure 4. ROC Curve of KAN-based Classification.

4.2. GenAI Roles Are on the Rise

As GenAI reshapes data-related occupations, identifying key skills linked to emerging roles is essential—not only for practitioners, but also for researchers seeking to understand the evolution of workforce demands. In response to *RQ₂*, the analysis examines how GenAI influences traditional data science workflows and identifies the specific skills most affected—highlighting the competencies that distinguish legacy roles from those aligned with GenAI-driven futures. This analysis supports alignment between workforce training and evolving industry needs. Figures 5 and 6 illustrate the top skills observed in *modern* and *traditional* (GenAI-aligned) job roles, respectively. These visualizations allow for a comparative analysis, highlighting the shift from statistical and reporting tools toward generative AI techniques such as prompt engineering, vector databases, and LLM frameworks. The next subsection discusses the implications of this transition and interprets the key differences using XAI methods.

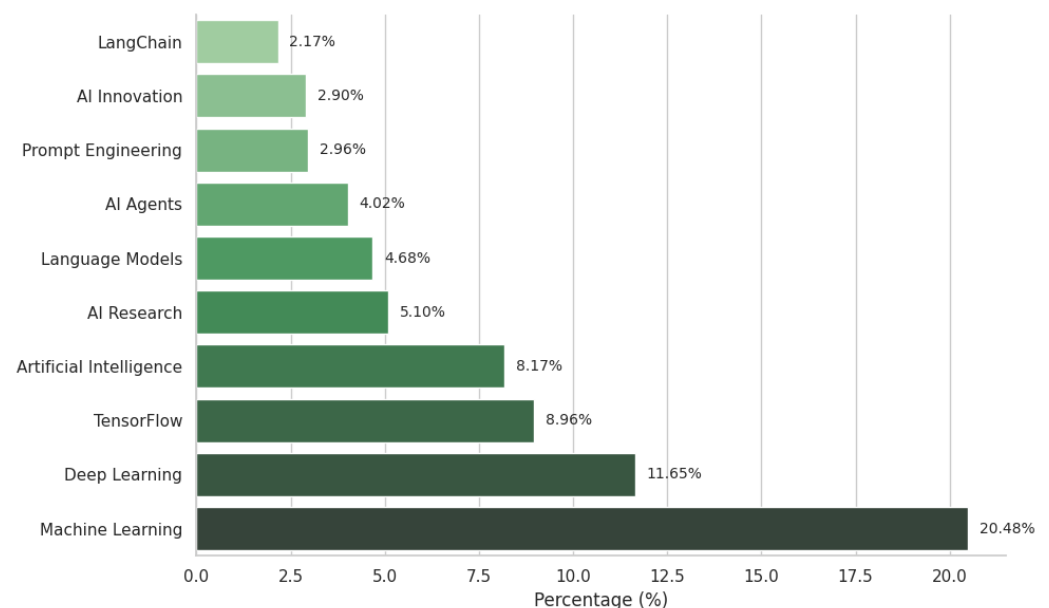


Figure 5. Top skills in GenAI (modern) job roles.

The rise of GenAI roles is reshaping the landscape of skill demand, as evidenced by the distribution of frequently appearing skills in modern job profiles. Figure 5 illustrates that Machine Learning dominates with 20.48% of occurrences, followed by Deep Learning

(11.65%) and TensorFlow (8.96%). These skills are among the most commonly listed in GenAI-related job advertisements, as shown in Figure 5. Moreover, skills like Artificial Intelligence (8.17%), AI Research (5.10%), and Language Models (4.68%) were among the most frequently mentioned in the dataset.

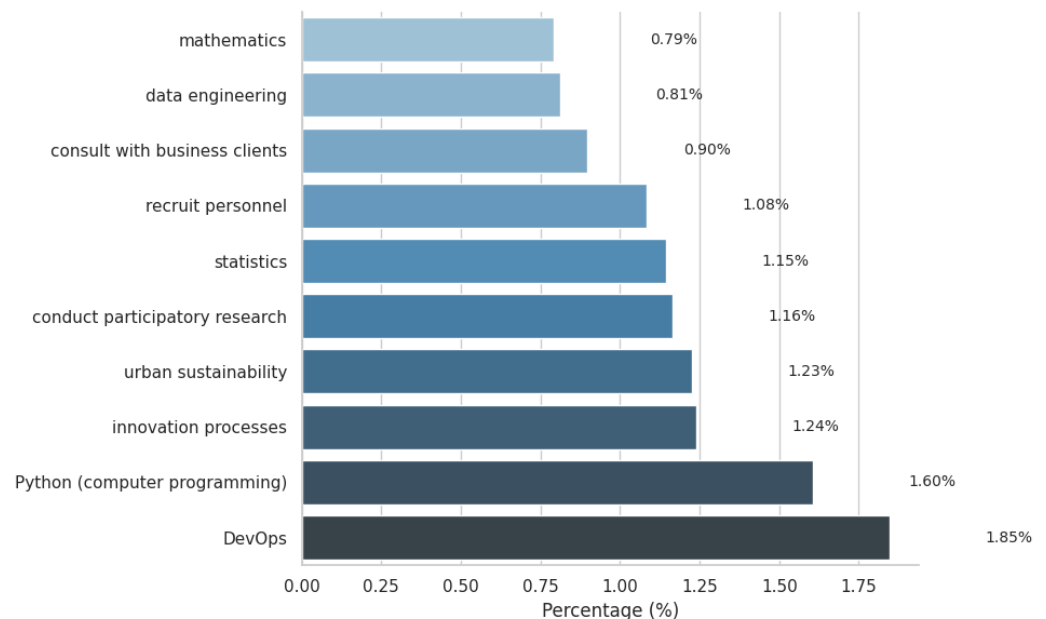


Figure 6. Top skills in traditional AI job roles.

In addition, emerging skills such as AI Agents (4.02%), Prompt Engineering (2.96%), AI Innovation (2.90%), and LangChain (2.17%) also appeared with notable frequency. The prominence of skills like Prompt Engineering reflects the increasing need for roles [86] that can interact effectively with LLMs, tailoring responses and optimizing system behavior. AI Agents refer to autonomous systems [87]—such as personal copilots or task managers—that leverage LLMs to perform complex, multi-step operations in business or technical environments. AI Innovation captures skills related to the design and deployment of novel GenAI use cases [88], including AI-driven product development, prototyping, and enterprise solutions. These skills are often not traditionally taught but are becoming essential in workflows that depend on generative models. This shift underscores a broader trend toward hybrid technical–cognitive competencies that blend programming literacy with human–AI interface design—indicating a strategic direction for both workforce upskilling and educational curricula in the near future.

On the other hand, Figure 6 presents the top skills most frequently associated with *traditional* job roles. Based on the standardized categorization provided by the ESCO taxonomy—enriched through our ESCOX tool—this distribution reflects the core competencies prevalent in non-GenAI domains. Among the most dominant skills are DevOps (1.85%) and Python (computer programming) (1.60%), which remain foundational for roles in software development and IT operations. Furthermore, interdisciplinary competencies such as innovation processes (1.24%), urban sustainability (1.23%), and conduct participatory research (1.16%) highlight the broader application of skills across policy, environment, and applied research contexts. Traditional domains continue to rely on analytical proficiencies like statistics (1.15%) and organizational capacities, including recruit personnel (1.08%) and consult with business clients (0.90%).

To illustrate the relationship between emerging and established skill categories, Figure 7 presents a Sankey diagram linking high-frequency GenAI-related skills (e.g., Prompt Engineering, Language Models) to their most co-occurring ESCO-classified coun-

terparts (e.g., DevOps, data science). Skill pairs were retained only if their co-occurrence exceeded a defined frequency threshold, ensuring meaningful associations.

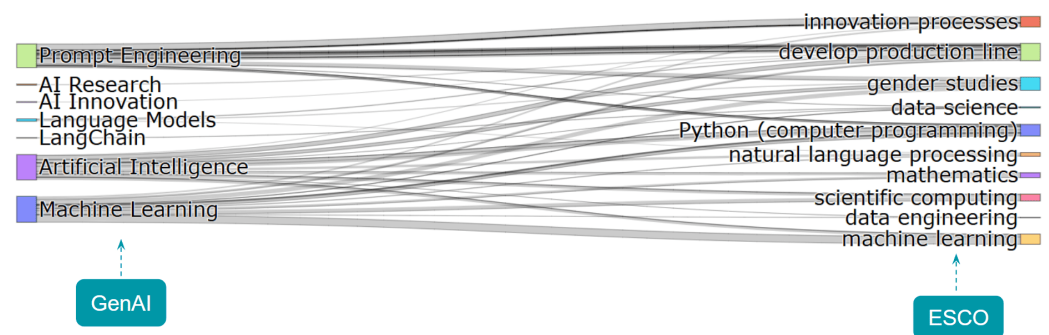


Figure 7. Interrelationships between GenAI and ESCO skills.

To guide the feature selection process for the KAN classifier, the top 70 most frequently occurring skills were selected as input features, as identified in our taxonomy classification table [89], which includes entries from both the GenAI and ESCO frameworks (see Table S4 in the Supplementary Materials). The top 70 most frequent skills were selected as model input features, based on their cumulative coverage and the inflection point in the frequency distribution curve. This cutoff captures approximately 86% of all skill mentions in the dataset, ensuring broad representativeness while minimizing overfitting caused by sparse, noisy, or infrequently observed features, and avoiding excessive computational cost and latency issues when running KANs with larger input dimensions. (see Supplementary Materials—Section S6.1 for empirical justification). In addition, empirical tests using thresholds from 40 to 100 showed that the top 70 skills offered the best trade-off between performance and semantic coverage. By incorporating both technical skills (e.g., LangChain, Prompt Engineering, MLflow) and transversal skills (e.g., communicate with stakeholders, develop creative ideas), the selected features maintain domain diversity and relevance. This curated subset enables the KAN model to effectively learn non-linear relationships without overfitting, thereby enhancing generalization and providing more interpretable outputs for classification tasks.

4.3. KANs Classification on Modern Job Roles

To address *RQ₃*, the analysis begins with a foundational question: *Can traditional data science roles be distinguished from their GenAI counterparts?* Before training the KAN model, a preliminary sanity check was conducted to assess the distributional variability of the input features. This step ensured that the extracted skill profiles carried sufficient discriminative power to meaningfully distinguish between GenAI-related and traditional roles—setting the stage for effective classification and interpretation. An extended analysis of SHAP value distributions across selected skills is provided in the Supplementary Material (Figure S4), illustrating variation in predictive contributions and supporting the robustness of the chosen model. To assess the robustness of SHAP-based feature importance analysis, we evaluated stability across multiple random seeds and under bootstrap resampling of the test set. When computing SHAP values with five different seeds (0, 42, 123, 777, and 999), the method consistently produced the exact same top 10 skills:

{Machine Learning, SAS Certified ModelOps Specialist, Language Models, Artificial Intelligence, TensorFlow, AI Agents, Python (computer programming), AI Research, data science, Deep Learning}

This resulted in a 100% overlap across seeds, indicating that the SHAP results are fully reproducible with respect to initialization variability. Bootstrap resampling of the test

set (five iterations, $n = 100$ each) yielded highly consistent outputs, with 7–8 of the top 10 skills preserved in each iteration. Variations were minor and typically occurred in the lower-ranked positions, demonstrating robustness to sampling fluctuations. Furthermore, we computed a base classification accuracy of 0.8130 for the KAN model on the test set and compared SHAP with a correlation-robust baseline via permutation importance. The overlap between SHAP (seed 0) and permutation importance rankings was 80%, confirming substantial agreement between these independent approaches. To enhance interpretability beyond SHAP rankings, we examined how the most influential skills affect the model's output through global effect plots. By visualizing Partial Dependence Plots (PDP) alongside Accumulated Local Effects (ALE) for the top-ranked skills, we capture both the overall and localized influence of each skill on the classification between modern and traditional roles. Supplementary Section S5.3—Figure S3 presents PDP and ALE plots for the most important skills as identified by the SHAP analysis.

The SHAP summary and decision plots presented in Figure 8 provide a granular interpretation of how individual skills influence the model's output in classifying modern versus traditional AI-related job roles. In these plots, negative SHAP values indicate a stronger association with traditional roles, while positive values are aligned with modern, GenAI-oriented classifications. Notably, Machine Learning and Deep Learning emerge as consistently high-impact features, serving as strong indicators for both role types depending on their contextual weight within each posting. The presence of *AI Agents*, *Language Models*, and *LangChain* is strongly associated with modern classifications, reinforcing their importance as frontier technologies shaping next-generation roles. On the other hand, skills such as *SAS Certified ModelOps Specialist* (based on the proprietary SAS platform, originally released in 1976 and continuously updated), *Qdrant*, and *DevOps* tend to cluster toward the traditional end, suggesting a continued reliance on established tools and infrastructure in conventional job settings. The decision plot further clarifies how accumulative contributions from specific skills guide the model toward its classification threshold, highlighting the linear and non-linear interactions captured by the KAN. These visualizations illustrate how individual skills contribute to the model's classification decisions and show the separation between features associated with traditional and GenAI-related roles.

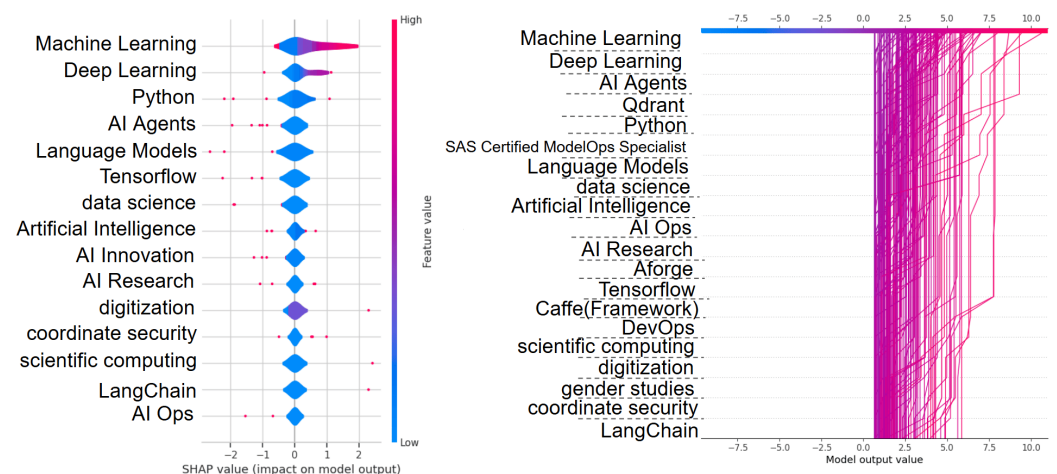


Figure 8. SHAP summary (left) and decision plot (right) focused on modern job roles.

Figure 9 illustrates the instance-level contributions of the most influential skills to the model's predictions for modern job roles, with the horizontal axis representing individual job instances, the vertical axis showing top contributing skills, and color intensity indicating each skill's SHAP value per instance.

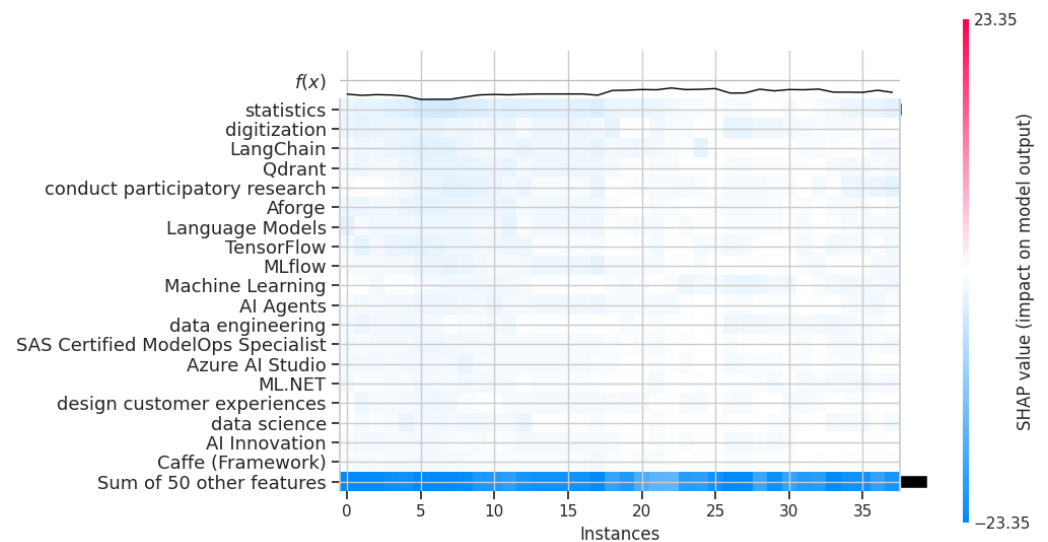


Figure 9. SHAP heatmap showing feature impact across individual instances for modern job roles. Darker red tones indicate stronger contributions to the model's prediction for the traditional class.

To illustrate local interpretability and validate the customized framework, SHAP values were examined for an AI/ML Data Scientist role that was correctly labeled as modern in the ground truth but assigned a relatively low prediction score of 0.4167 by the model. Figure 10 presents a SHAP waterfall plot, revealing that the skills AI Agents, Python (computer programming), communicate with customers and Prompt Engineering contributed most positively to the classification output. Notably, the identified skills are frequently associated with GenAI-related occupations, as reflected in the model's classification outputs. However, the presence or absence of other influential skills diluted the total activation, resulting in a conservative prediction.

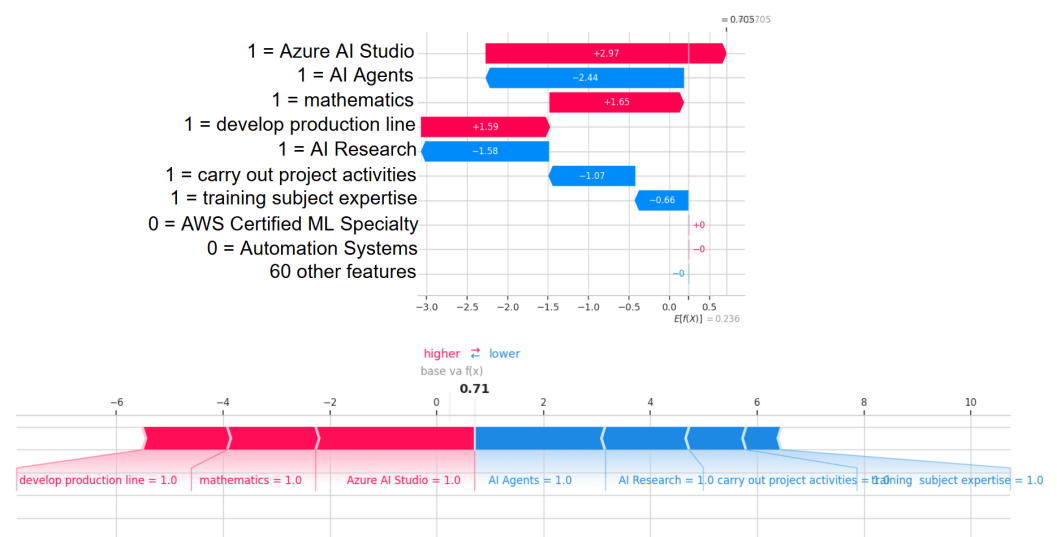


Figure 10. SHAP visualization showing key skills that affect the likelihood of classifying a job as modern—AI Deployment Engineer.

Furthermore, Figure 10 complements this with a force plot that visualizes the same cumulative effect, with red bars pushing the prediction toward the modern class and blue indicating suppression from inactive or missing GenAI-related skills. The most influential positive contribution comes from Azure AI Studio with a SHAP value of +2.97, followed by mathematics and develop production line with values of +1.65 and +1.59, respectively. These features collectively push the prediction significantly toward the modern

classification, reflecting the emphasis on cloud-based AI tooling, quantitative foundations, and real-world AI deployment capabilities in such roles. On the other hand, certain features exert a downward influence on the output. Notably, AI Agents (−2.44), AI Research (−1.58), and carry out project activities (−1.07) contribute to lowering the score. These features are commonly associated with academic or research-oriented skill sets, as observed in the classification patterns. Despite these negative contributions, the overall prediction remains firmly in favor of the modern class. This example highlights how SHAP allows us to dissect the decision boundary, revealing how modern roles are distinguished not only by frontier AI capabilities but also by their alignment with operational AI infrastructure and automation tools. For further information related to the XAI interpretation of modern job roles, please refer to the Supplementary Materials—Section S6.2.

4.4. KANs Classification on Traditional Job Roles

After examining how emerging roles align with GenAI-driven transformations, the focus shifts to the foundations upon which they build: traditional occupations. These roles continue to anchor much of the labor market and deserve equally rigorous analysis. A detailed SHAP-based interpretation of their classification is provided in the Supplementary Materials (Figure S5).

To uncover the skill signals that define traditional job classifications, SHAP summary and decision plots specific to non-GenAI occupations were examined, as depicted in Figure 11. These visualizations illustrate how the model’s decisions are influenced by a diverse combination of legacy certifications, classical data science knowledge, and general-purpose competencies. In particular, *SAS Certified ModelOps Specialist*, *Qdrant*, and *Machine Learning* consistently appear as key drivers in model predictions, confirming their anchoring role in traditional AI practices. Additionally, soft skills such as *communication*, *think creatively*, and *English* also contribute meaningfully, highlighting the hybrid expectations embedded in contemporary traditional roles. The decision plot emphasizes how different combinations of features accumulate toward a classification outcome, with visible variance in feature paths reinforcing the model’s ability to handle non-linearity and mixed-domain input.

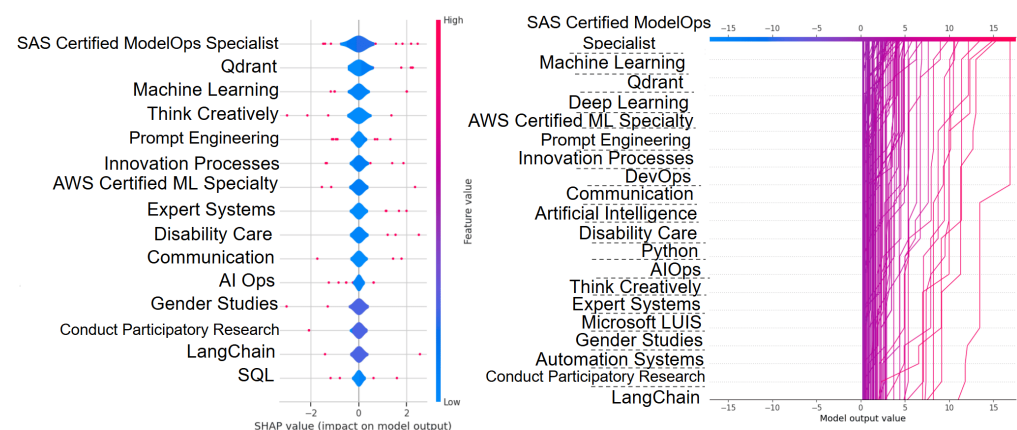


Figure 11. SHAP summary (left) and decision plot (right) focused on traditional job roles.

It is also important to note the prevalence of lower (blue or near-zero) SHAP values across most features and instances. The SHAP visualizations indicate that only a limited number of skills per job posting have strong positive contributions to the classification outcome. This likely reflects the fact that most postings contain a small subset of the full skill lexicon. This reinforces the interpretability advantage of SHAP, as it transparently

captures how a few discriminative skills, rather than the full feature set, dominate the model's predictive logic.

Figure 12 displays a SHAP heatmap depicting per-instance feature contributions for traditional job role classifications. *Machine Learning* and *SAS Certified ModelOps Specialist* consistently show strong positive SHAP values, while features such as *AI Agents* and *Language Models* exhibit sparse activation, aligning with their limited relevance in traditional roles.



Figure 12. SHAP heatmap showing feature impact across individual instances for traditional job roles. Darker red tones indicate stronger contributions to the model's prediction for the traditional class.

To further illustrate model interpretability, a representative example of a traditional data science role—*Data Quality Analyst*—was presented and its feature attributions are analyzed using the SHAP decision plot shown in Figure 13. The SHAP visualization decomposed the model's prediction into individual skill contributions. In this case, the prediction $f(x)$ is approximately 0.04, significantly lower than the average model output $E[f(x)] = 0.387$, indicating that the model confidently classifies this instance as a traditional role.

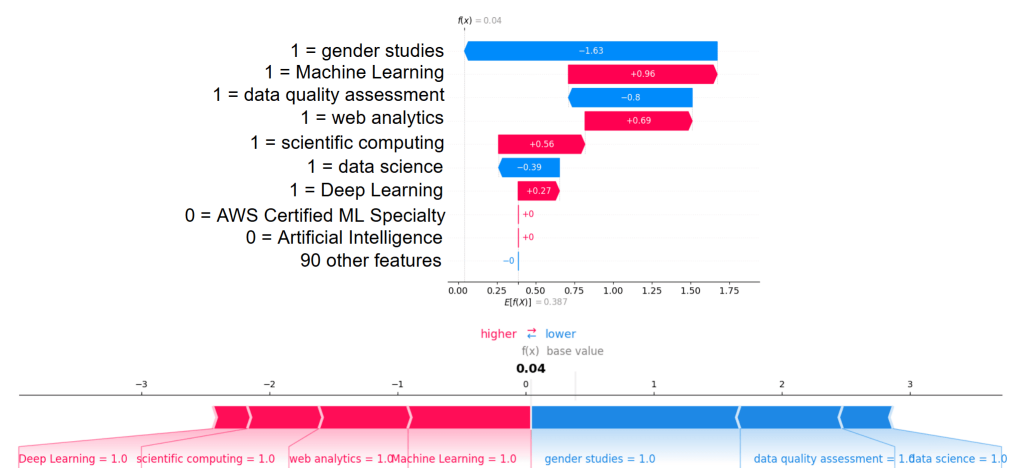


Figure 13. SHAP visualization showing key skills that affect the likelihood of classifying a job as traditional—*Data Quality Analyst*.

Skills such as *gender studies*, *data quality assessment*, and *data science* exerted negative SHAP values, pulling the model's prediction downward toward the traditional class. The most influential downward force came from *gender studies* (−1.63), followed by *data quality assessment* and *data science*, suggesting that these skills are strongly aligned with the traditional job type in the model's learned representation. Conversely, upward contributions are made

by *Machine Learning*, *web analytics*, and *scientific computing*, which slightly nudged the output toward modern classification but are not dominant enough to shift the final decision. The SHAP visualizations showed the contribution of both technical and contextual skill features to the model's classification. In the presented example, the inclusion of traditional skills resulted in a low classification score, indicating alignment with the traditional job role category.

Importantly, several traditional skills identified by the model—such as *Machine Learning*, *scientific computing*, and *web analytics*—represent foundational capabilities that are now being extended in GenAI contexts. These competencies are increasingly recontextualized as prerequisites for advanced GenAI workflows, such as training or fine-tuning LLMs, developing AI pipelines, or building agent-based architectures. As GenAI adoption accelerates, the transition from traditional to modern roles may not always require a radical skill shift, but rather an upskilling trajectory built upon these existing capabilities. This insight has implications for both workforce planning and curriculum design, encouraging a more fluid reskilling path rather than a disruptive reorientation. For further information related to the XAI interpretation of traditional job roles, please refer to the Supplementary Materials—Section S6.3.

5. Practical Implications

As professionals navigate a labor market reshaped by GenAI, distinguishing between traditional and emerging roles has become essential for informed career and training decisions. Figure 14 illustrates the study's framework—a structured, data-driven pipeline designed to classify and interpret OJA based on emerging GenAI-related skills, addressing *RQ₄*: *How can real-time skill explainability be used to support stakeholders in identifying GenAI readiness and guiding upskilling decisions?*. The process begins with real-world job listings from EU platforms, where skill data are extracted via ESCOX and a GenAI lexicon. These are structured and fed into a KAN classifier that distinguishes traditional from GenAI-relevant roles. Using XAI techniques, the skills influencing each classification were identified, offering greater transparency and trust in the model's decisions. Beyond classification, the insights generated by our framework could be leveraged to support real-time labor market interventions by informing policy design, shaping targeted reskilling programs, and aligning educational curricula with rapidly evolving workforce demands.

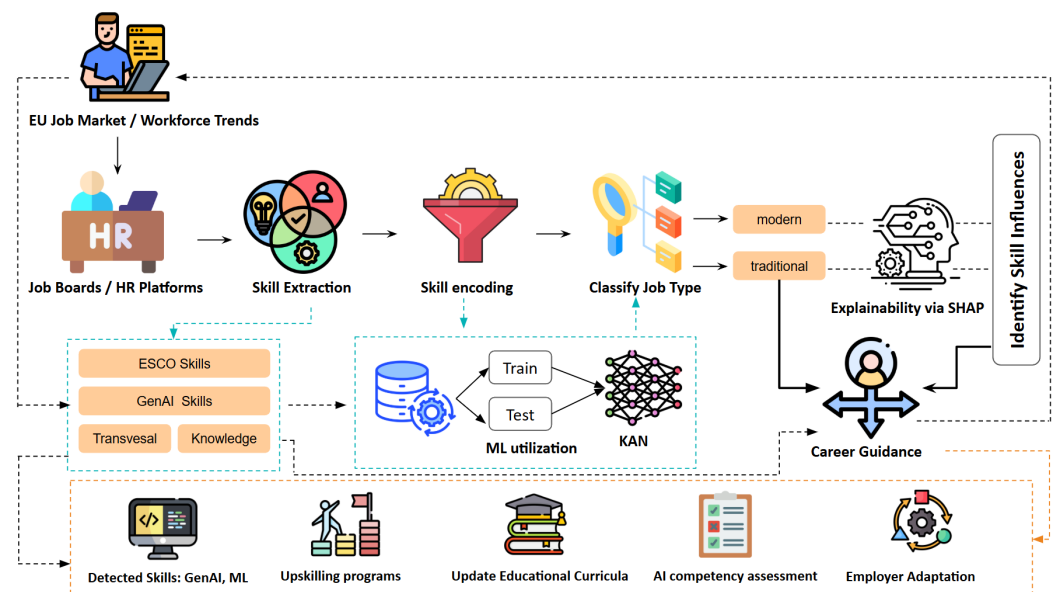


Figure 14. A data-driven and XAI framework for classifying job types.

Curriculum Development and Reform. The outcomes of our explainable job classification framework have direct implications for educational curriculum design. By identifying which GenAI-related skills (e.g., Prompt Engineering, AI Agents, LangChain) are most predictive of modern OJA, our model offers a data-driven foundation for revising academic programs. Educational institutions across Europe, such as *ETH Zurich's* (<https://ethz.ch/en/the-eth-zurich/education.html> (accessed on 19 August 2025)) and *Delft University of Technology* (<https://www.tudelft.nl/en/> (accessed on 19 August 2025)), can integrate these competencies into computer science, data science, and engineering curricula to better align with labor market needs. For instance, the *ETH Zurich Master in Data Science* program (<https://ethz.ch/en/studies/master/degree-programmes/engineering-sciences/data-science.html> (accessed on 19 August 2025))—offered collaboratively by the Departments of Computer Science, Mathematics, and Information Technology & Electrical Engineering—already includes elective modules such as “Natural Language Understanding” and “Reliable and Interpretable Artificial Intelligence”. A dedicated module such as “Prompt Engineering for LLMs” could be integrated within this framework as a new elective, either as an augmentation to the existing NLP track or as part of the AI specialization stream.

Targeted Upskilling & Reskilling Programs. The insights derived from our classification framework can inform the development of targeted upskilling and reskilling initiatives, especially for professionals in occupations classified as “traditional.” By pinpointing the specific GenAI-related skills absent in these roles—such as Prompt Engineering, LLMs, or LangChain—the model offers a skill gap diagnosis that can guide the design of modular and personalized training programs (Figure 5). Public employment services, vocational training providers, and private EdTech platforms could use these insights to tailor interventions that address concrete labor market needs. This is particularly valuable for sectors or job profiles at risk of technological obsolescence, “where reskilling efforts must align with the rapidly evolving AI landscape. The use of XAI ensures that recommended upskilling pathways are not only aligned with labor market trends but are also interpretable and justifiable to learners, employers, and policymakers alike. For example, companies like *PwC* (<https://www.pwc.com/gr/en.html> (accessed on 19 August 2025)) and *EY* (https://www.ey.com/en_gl (accessed on 19 August 2025)) could offer targeted GenAI upskilling programs, such as a “Generative AI Certification for Data Analysts,” focusing on tools like LangChain and prompt design, to support professionals transitioning from legacy analytics roles. This industry-led upskilling effort complements academic initiatives, highlighting a dual-track approach to workforce development. Notably, this approach is fully aligned with the strategic objectives of EU-funded projects such as SKILL-LAB [90,91], which aim to monitor skill demand and supply in real time and reduce the structural mismatches currently observed in the European labor market.

AI Competency Assessment. Our framework integrates XAI techniques into automated skill diagnostics, offering employers actionable insights into employee readiness for AI-centric roles. Using SHAP explanations at the individual prediction level (Figure 8), organizations can identify which GenAI-related skills contribute to or are missing from an employee’s profile. This supports the development of interactive dashboards visualizing personalized skill gaps and readiness levels, aiding strategic workforce planning and targeted professional development. The interpretability of SHAP ensured transparency in assessments, enabling data-driven decisions for upskilling and internal mobility.

AI-Driven Career Guidance. The explainable outputs of our classification system offer practical value for personalized job matching. Using SHAP-based interpretations (Figure 13), the model highlights skills that influence the classification of job roles as modern or traditional. For example, data science bootcamps such as those offered by *Le Wagon*

(<https://www.lewagon.com/> (accessed on 19 August 2025)), can incorporate hands-on modules using LangChain to build domain-specific chatbots and LLM-driven pipelines for data querying and summarization. This transparency enables individuals and career advisors to identify specific skill gaps and recommend relevant GenAI competencies—such as MLOps or Prompt Engineering—to improve alignment with modern job demands. Such skill-aware insights support a more targeted, adaptable approach to career planning in the evolving AI-driven labor market.

Employer Adaptation. As technology evolves, employers must align roles with GenAI labor demands. Our framework enables organizations to assess whether their postings reflect emerging competencies. By identifying missing skills—like LangChain, AI Agents, or Prompt Engineering—SHAP-based explanations guide updates to role definitions, hiring criteria, and training. This supports a feedback loop for employers to adapt recruitment and upskilling strategies, fostering readiness for GenAI integration.

To facilitate actionable, skill-based insights for HR professionals, we developed the KANVAS JOB ANALYZER (Figure 15), a core component of the KANVAS framework, featuring a Python backend and Gradio-powered frontend, provides an intuitive, multi-step workflow designed for HR professionals to explore, interpret, and act on job-level skill insights. The interface components are annotated as follows: Step ① allows HR managers to select a job index from the dataset using an intuitive slider. Each index corresponds to a unique job posting, which initiates the analysis pipeline. Step ② displays the system’s prediction of whether the selected role is modern or traditional, supported by a confidence score and a generated job title and description based on the most influential skills. Step ③ highlights the top contributing skills using a ranked textual explanation derived from gradient-based SHAP analysis, helping HR professionals understand the model’s reasoning. Step ④ presents a radar plot that visualizes the importance of the top six skills, offering a compact and comparative overview that supports decision-making around training and reskilling initiatives.

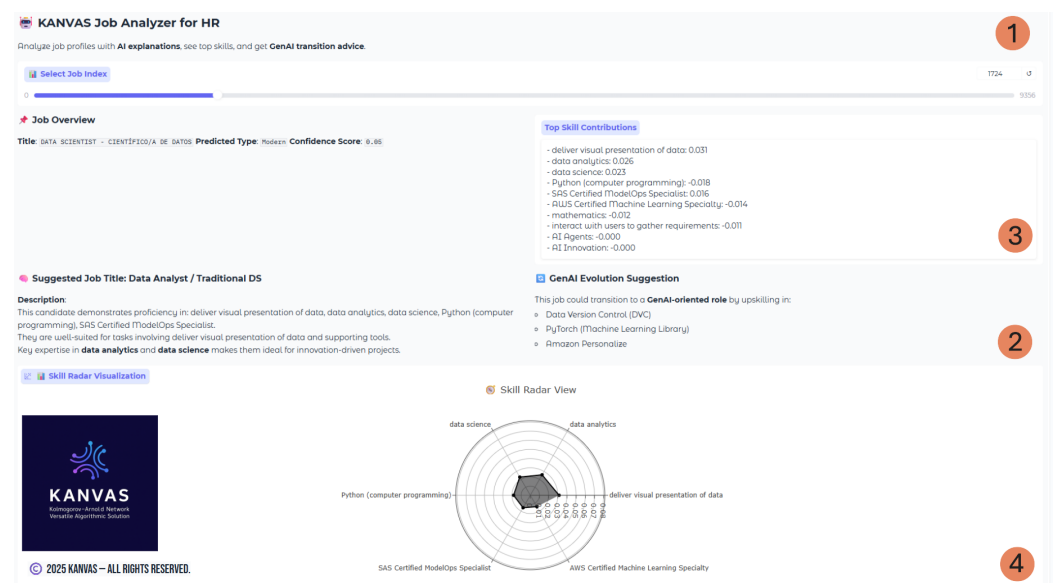


Figure 15. KANVAS Job Analyzer.

6. Discussion

In response to RQ_4 , which explores how real-time skill explainability can support stakeholders in identifying GenAI readiness and guiding upskilling decisions, our framework integrates XAI techniques to make skill classification more transparent and actionable. By integrating XAI techniques—specifically SHAP and the KAN—our framework enhances

transparency in skill-based classification. This combination enables interpretable insights that are particularly valuable in labor market applications, where understanding the rationale behind classification is critical. SHAP values provide fine-grained attributions of skill influence for each prediction, supporting both local (per-job) and global (dataset-wide) analysis. Such transparency is especially beneficial for stakeholders such as career advisors and curriculum designers, who can identify specific skill gaps and update training pathways accordingly. This explainable and actionable approach enables more informed, human-centric decision-making in response to evolving workforce demands driven by GenAI technologies.

As GenAI technologies become embedded in everyday work environments, they risk deepening existing inequities in job access—particularly for individuals and regions with limited exposure to foundational AI skills. The transition toward AI-augmented roles may disproportionately benefit workers in urban or well-resourced areas, while leaving behind communities lacking digital infrastructure, training opportunities, or educational support. This uneven distribution of GenAI competencies can exacerbate the digital divide, reinforcing systemic disparities in employment and career advancement. To address these emerging skill gaps, targeted regional policy interventions are proposed. National and EU-level agencies like Agency for Support for BEREC (BEREC Office) (https://european-union.europa.eu/institutions-law-budget/institutions-and-bodies/search-all-eu-institutions-and-bodies/agency-support-berec-berec-office_en (accessed on 19 August 2025)) should prioritize funding for localized GenAI training hubs in rural and economically lagging regions, aligned with labor market analytics. Public–private partnerships can support mobile training units, community colleges, and microcredentialing platforms focused on high-impact GenAI skills. Our framework, by making skill requirements transparent and traceable, offers a tool for addressing this challenge. As illustrated in Figure 9, SHAP-based explainability reveals which GenAI-related skills most strongly influence job classification outcomes. By identifying where GenAI-related skills are most in demand—and which are most predictive of access to emerging roles—it becomes possible to design more inclusive reskilling programs, tailored to underserved populations.

Figure 12 offers valuable insights into how traditional job roles are shaped by specific skill indicators within the model’s decision-making process. The consistently high SHAP values for Machine Learning, SAS Certified ModelOps Specialist, and TensorFlow affirm the dominance of structured, certifiable, and classical AI-related skills in legacy job classifications. The weak or absent influence of emerging competencies such as AI Agents and Language Models further delineates the conceptual boundary between traditional and GenAI-driven roles. The relatively flat $f(x)$ trajectory across instances suggests classification stability, indicating that traditional roles are more semantically uniform than their GenAI counterparts. This semantic stability, while offering reliability, may also signal stagnation if not addressed through policy-driven reskilling initiatives. To prevent obsolescence, regional and sectoral policymakers should launch upskilling programs that bridge traditional skills with GenAI competencies—for instance, certifying current data analysts in LLM usage, prompt design, or agent orchestration. A concrete example is the “AI Forward” initiative by Dapra Innovation office (<https://www.darpa.mil/research/programs/ai-forward> (accessed on 19 August 2025)), which offers GenAI training for legacy tech staff. These transitions can be supported by modular training incentives or GenAI-readiness grants targeting legacy professionals. From a policy and curriculum perspective, this pattern highlights the inertia in traditional role definitions and the potential risk of obsolescence unless reskilling interventions are proactively implemented. This separation of skill sig-

natures is not only a model finding but also a reflection of a real-world divide in digital workforce evolution.

While our framework effectively distinguishes between traditional and GenAI-related roles, it also uncovers the emergence of *hybrid job profiles*, identified through insights from the KANVAS Job Analyzer (Figure 15)—roles that combine classical data science foundations with GenAI competencies like prompt engineering and LLM integration. These blended profiles signal a labor market in rapid evolution, outpacing current taxonomies and posing alignment challenges. *HR professionals* can use the analyzer to benchmark job ads against industry trends, identify missing GenAI skills in job descriptions, and design adaptive upskilling paths. For example, a hiring manager could detect the underrepresentation of GenAI skills in current roles and update internal role definitions accordingly. *Policymakers* might leverage the underserved skill zones to prioritize funding for regional retraining programs, especially in sectors lagging behind (e.g., manufacturing or public service). *Educational institutions* can support this shift by embedding GenAI into legacy STEM modules and offering micro-credentials focused on LLMs and prompt design. To operationalize this hybrid classification, an LLM-based pipeline was employed for job labeling. The LLM's semantic reasoning over job titles and descriptions enhanced detection of nuanced cases beyond rule-based systems. Still, its deployment must be approached with caution due to issues like *hallucinations*, *prompt instability*, and *model bias*—factors that demand expert oversight, particularly when informing public policy, training design, or career guidance infrastructure.

Our findings underscore critical limitations in current skill taxonomies such as ESCO. While comprehensive for established occupations, these frameworks struggle to accommodate the rapid emergence of GenAI competencies. This shortfall—further explored in Section 3.1—represents a core issue our study aims to address. Key skills frequently extracted from OJA data, including *LLM fine-tuning*, *retrieval-augmented generation (RAG)*, and *prompt optimization*, are often missing or insufficiently represented in official classifications. This mismatch between real-world demand and taxonomy coverage creates downstream risks: public employment services may misalign reskilling pathways, educational providers might overlook critical content, and labor market analytics could underrepresent evolving domains. To maintain relevance, skill taxonomies must adopt an agile update cycle—leveraging data-driven techniques such as large-scale OJA analysis and explainable AI to iteratively integrate emergent skills. Without such reform, static frameworks risk fossilizing the labor market's view of occupational needs, leaving job seekers underinformed and policymakers misdirected. Our contribution, by linking transparent classification models with skill signal extraction, provides a methodological blueprint for bridging this taxonomy gap in a dynamic and interpretable way.

The practical utility of our classification framework extends beyond academic inquiry, offering actionable value to policymakers, workforce planners, and training providers, as detailed in Figure 15. Through interpretable SHAP-based outputs, decision-makers can identify skill gaps and prioritize targeted upskilling pathways. The dashboard developed in our study supports real-time monitoring of labor market trends, enabling interventions such as curriculum reform and career transition programs. For public services, it enhances career guidance; for enterprises, it informs strategic workforce planning. Beyond data science, the methodological structure of KANVAS holds promise for adaptation across other sectors impacted by GenAI—such as healthcare, legal technology, and education—where emerging AI tools increasingly intersect with domain-specific expertise. By refining skill lexicons and taxonomies, our pipeline can be generalized to classify evolving job roles and track AI-driven skill shifts across diverse industries.

As shown in Table 3, the model demonstrates higher recall for modern roles but higher precision for traditional ones. This trade-off may reflect structural differences in how these roles are represented in the data: modern roles often involve broader and less standardized skill sets, increasing the chance of false positives, whereas traditional roles rely on more consistent and narrowly defined skills, improving precision but potentially overlooking hybrid or emerging profiles. The observed imbalance between precision and recall across classes can be attributed to two main factors. First, several features—such as machine learning or data visualization—are common to both modern and traditional roles, leading to class confusion when shared skills dominate the input vector. This results in overprediction of the modern label, producing high recall but lower precision for that class. Second, the dataset contains a higher number of traditional job postings, which may introduce representational bias and limit the model’s ability to generalize across the more diverse and evolving patterns found in modern roles.

6.1. Generalizability and Limitations

6.1.1. Internal Validity

Several factors affect the internal validity of our framework and its interpretation reliability. First, the GenAI lexicon—manually constructed by the authors to comprehensively capture emerging GenAI-related competencies—may introduce subjective bias and risks overlooking less adopted or newly emerging terms. Additionally, while SHAP values provide transparency, they are approximations that can be sensitive to feature correlations and input noise, particularly in high-dimensional skill spaces. The model’s performance is also contingent on the quality of labeled data. Although KANs allow flexible non-linear modeling, they may underperform in cases where skills are inconsistently expressed across job descriptions or where signal sparsity reduces clarity. Furthermore, the semantic overlap between some skills led to job postings being annotated as both traditional and GenAI-related (as noted in Section 3.2), which, although expected, underscores the hybrid nature of many roles—a central motivation of this study.

6.1.2. External Validity

In terms of generalizability, several limitations constrain external validity of the proposed framework. The training dataset reflects a curated snapshot of job advertisements focused on traditional and GenAI roles, which does not fully represent the complexity and breadth of the wider EU labor market. This limitation, driven by computational constraints in processing large-scale data with KANs, restricts the framework’s scalability. Importantly, the scope of our dataset is primarily limited to sectors heavily represented in OJA—such as data science and emerging GenAI roles. Furthermore, while the dataset predominantly includes postings from EU-based platforms (e.g., LinkedIn, Kariera), labor market dynamics and skill taxonomies may vary across global regions. As such, the generalizability of findings to non-EU markets or to sectors such as healthcare, education, or public administration should be approached with caution. Additionally, the selected job postings are primarily drawn from platforms focused on technical and AI-heavy roles, resulting in limited representation from sectors such as management or the humanities. This skews the model’s relevance when applied to non-technical or cross-sector occupations. Furthermore, the use of predefined taxonomies like ESCO and a manually curated GenAI lexicon may overlook recent skill trends. As such, successful deployment of the system requires context awareness and regular updates to ensure applicability across varied industries and regions. A key limitation of our curriculum reform suggestions lies in the assumption of uniform institutional readiness. In practice, educational institutions differ widely in their capacity to integrate GenAI competencies—due to disparities in infrastructure, faculty

expertise, governance structures, and access to up-to-date tools or datasets. As a result, the proposed curriculum adaptations may be more feasible for well-resourced universities in urban or research-intensive settings, while posing significant implementation barriers for underfunded or smaller institutions. We acknowledge that automated classification systems may lead to misclassification risks with potential downstream impacts on candidate screening. In this study, a human annotation protocol was used to validate the classification outputs, but for any real-world operational deployment, we recommend implementing the framework in a human-in-the-loop setting to ensure oversight and mitigate potential harm.

6.1.3. Privacy and Policy

While the dataset used in this study was collected exclusively from publicly accessible job boards (LinkedIn and Kariera) and did not include any PII, limitations inherent to secondary data usage remain. Kariera's Privacy Policy and Terms of Service explicitly frame data handling in accordance with GDPR requirements [92], and LinkedIn's public-facing Data Privacy Notice demonstrates transparent data practices adhering to European data protection principles [93]. Furthermore, industry guidance for GDPR-compliant recruitment underscores that responsibly accessed public data can be legally collected under such frameworks [94]. Nonetheless, as data availability and legal interpretations evolve, future work should continue to monitor compliance practices and ethical considerations in automated job data analysis.

7. Conclusions & Future Work

In conclusion, KANVAS was introduced as a skill-based job classification framework designed to distinguish between traditional and GenAI-oriented roles using OJA. Our approach offers an interpretable method for uncovering key skill signals and mapping their influence on job categorization, particularly through the use of explainable AI techniques. The explainability layer helped clarify model predictions, offering preliminary insights that may support decision-making in education, workforce planning, and policy contexts.

Future research may explore multi-class or hybrid classification settings, combining both traditional and GenAI skill sets into composite job roles. Additionally, a promising direction involves exploring career pathway modeling and reinforcement learning as conceptual tools for simulating skill transitions and developing personalized upskilling strategies. While preliminary, these ideas point to potential enhancements that could make KANVAS more adaptive and responsive to evolving labor market dynamics.

To mitigate the observed imbalance between precision and recall, as detected in Table 3, future work should consider incorporating richer contextual features into the input space—such as job seniority level, industry sector, or skill co-occurrence frequency—to better distinguish overlapping competencies between modern and traditional roles. Additionally, ensemble learning techniques (e.g., stacking with interpretable models) or cost-sensitive loss functions could help optimize for more balanced performance across both classes.

Moreover, the KANVAS framework could be extended to enable longitudinal analyses that track temporal shifts in job structures and evolving skill requirements, for instance by applying time-stamped OJA data across quarterly or yearly intervals. These enhancements may support early identification of disruptive trends—such as the rapid adoption of LLM-based workflows—or the decline of legacy roles. Additionally, incorporating region-sensitive skill mapping using location-tagged job postings (e.g., NUTS-2 regions in the EU) could help quantify geographic disparities in GenAI preparedness, offering input for more targeted policy strategies such as region-specific upskilling programs or GenAI readiness grants. Integrating curriculum alignment mechanisms—linking skill

gaps found through KAN-based predictions to relevant educational content—may also assist institutions in improving alignment between training and market needs. While these expansions could enhance KANVAS as a responsive and transparent decision-support system, we recognize that broader testing across diverse sectors and datasets is essential before drawing generalized conclusions. Future work could explore ablation experiments comparing model performance with and without ESCO-aligned skill mappings to better quantify their contribution to classification accuracy.

While it is clearly established that a dichotomy between traditional AI and GenAI exist—and numerous studies validate this claim [39–43]—there remains potential for extending the present formulation into a multi-label classification framework. Such an approach could incorporate additional categories beyond the two used in this study, for example: (i) hybrid roles that combine both traditional and GenAI competencies, (ii) automation-augmented roles where AI complements but does not replace human expertise, and (iii) emerging interdisciplinary roles that blend AI capabilities with domain-specific knowledge such as law, medicine, or education. Furthermore, while we rely on established skill taxonomies such as ESCO and Lightcast, alternative approaches—such as topic modeling—could be employed to uncover latent themes or skill clusters present in job postings.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/a18090554/s1>. Additional details regarding the tools employed, the skill extraction procedures, model formalizations, and extended analyses are provided in the Supplementary Materials.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
GenAI	Generative Artificial Intelligence
LLM	Large Language Model
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
ESCO	European Skills, Competences, Qualifications, and Occupations
ISCO	International Standard Classification of Occupations
KAN	Kolmogorov–Arnold Network
ML	Machine Learning
CV	Curriculum Vitae

RAG Retrieval-Augmented Generation
 CSV Comma-Separated Values

Appendix A

Table A1. The list of GenAI skills.

GenAI Skills	
Artificial Intelligence Development	DALL-E Image Generator
Artificial Intelligence Risk	CrewAI
Artificial Intelligence Systems	Azure OpenAI
Artificial General Intelligence	AutoGen
Artificial Neural Networks	Image Captioning
AI/ML Inference	Image Inpainting
Applications of Artificial Intelligence	Image Super-Resolution
AI Agents	Natural Language Generation (NLG)
AI Alignment	Large Language Modeling
AI Innovation	Language Models
AI Research	Natural Language Understanding (NLU)
AI Safety	Natural Language User Interface
Attention Mechanisms	LangChain
Adversarial Machine Learning	Langgraph
Agentic AI	Microsoft Copilot
Agentic Systems	Microsoft LUIS
Autoencoders	Prompt Engineering
Association Rule Learning	Retrieval Augmented Generation
Activity Recognition	Sentence Transformers
3D Reconstruction	Operationalizing AI
Backpropagation	Supervised Learning
Bagging Techniques	Unsupervised Learning
Bayesian Belief Networks	Transfer Learning
Boltzmann Machine	Zero Shot Learning
Classification and Regression Tree (CART)	Soft Computing
Deeplearning4j	Sorting Algorithm
Concept Drift Detection	Training Datasets
Deep Learning	Test Datasets
Deep Learning Methods	Test Retrieval Systems
Deep Reinforcement Learning (DRL)	Dlib (C++ Library)
Computational Intelligence	Topological Data Analysis (TDA)
Convolutional Neural Networks	Swarm Intelligence
Cognitive Computing	Spiking Neural Networks
Collaborative Filtering	Variational Autoencoders
Ensemble Methods	Sequence-to-Sequence Models (Seq2Seq)
Expectation Maximization Algorithm	Transformer (Machine Learning Model)
Expert Systems	Stable Diffusion
Federated Learning	Small Language Model
Few Shot Learning	Apache Mahout
Gradient Boosting	Apache MXNet
Gradient Boosting Machines (GBM)	Apache SINGA
Hidden Markov Model	Aforge
Incremental Learning	Amazon Forecast
Inference Engine	Azure OpenAI
Hyperparameter Optimization	ChatGPT
Fuzzy Set	DALL-E Image Generator
Genetic Algorithm	CatBoost (Machine Learning Library)
Genetic Programming	Chainer (Deep Learning Framework)

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